

# Series & Binomial Theorem



## Arithmetic sequences

- 1 An arithmetic sequence has first term  $a = 5$  and common difference  $d = 3$ . Find the 20th term.
  - 2 Find the sum of the first 15 terms of the arithmetic sequence 4, 7, 10, 13, ...
  - 3 An arithmetic sequence has 12th term 47 and common difference  $d = 4$ . Find the first term  $a$ .
  - 4 The 5th term of an arithmetic sequence is 23 and the 9th term is 39. Find the common difference  $d$  and the first term  $a$ .
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## Geometric sequences

- 5 A geometric sequence has first term  $a = 3$  and common ratio  $r = 2$ . Find the 10th term.
  - 6 Find the sum of the first 8 terms of the geometric sequence 2, 6, 18, 54, ...
  - 7 A geometric sequence has first term  $a = 4$  and common ratio  $r = \frac{1}{2}$ . Find the sum of the first 6 terms.
  - 8 A geometric sequence has 2nd term  $u_2 = 6$  and 5th term  $u_5 = 48$ . Find the common ratio  $r$  and the first term  $a$ .
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## Sum to infinity of a geometric series

- 9 Find the sum to infinity of the geometric series  $8 + 4 + 2 + 1 + \dots$
  - 10 A geometric series has  $S_\infty = 45$  and first term  $a = 9$ . Find the common ratio  $r$ .
  - 11 Find the sum to infinity of the geometric series  $20 - 10 + 5 - 2.5 + \dots$
  - 12 A geometric series has first term  $a = 12$  and sum to infinity  $S_\infty = 16$ . Find the common ratio  $r$ .
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## Sigma notation

- 13 Evaluate  $\sum_{k=1}^5 (2k + 1)$ .
- 14 Evaluate  $\sum_{k=1}^6 (3k - 2)$ .

- 15 Evaluate  $\sum_{k=1}^4 k^2$ .
- 16 Evaluate  $\sum_{k=3}^7 (2k - 1)$ .

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### Binomial expansion for positive integer powers

- 17 Use the binomial theorem to expand  $(x + 2)^4$  in ascending powers of  $x$ .
- 18 Find the coefficient of  $x^3$  in the expansion of  $(2x - 1)^5$ .
- 19 Find the coefficient of  $x^2$  in the expansion of  $(1 + 3x)^6$ .
- 20 Find the coefficient of  $x^4$  in the expansion of  $(x - 2)^6$ .

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### Using Pascal's triangle and factorial notation

- 21 Evaluate  $\binom{6}{2}$  using  $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ .
- 22 Evaluate  $\binom{7}{3}$  using  $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ .
- 23 Evaluate  $\binom{8}{5}$  using  $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ .

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### Applications of arithmetic and geometric series

- 24 A theater has 20 seats in the first row, and each subsequent row has 2 more seats than the row before. Find the total number of seats in the first 25 rows.
- 25 A stack of logs has 30 logs on the bottom row, with each row above having 2 fewer logs than the row below, down to a top row of exactly 2 logs. Find the total number of logs in the stack.
- 26 A ball is dropped from a height of 4 meters, and each bounce reaches 60% of the height of the previous bounce. Find the total distance the ball travels after infinitely many bounces, using the sum to infinity of the geometric series of bounce heights.

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### A well-designed word problem: a savings account with compound growth

- 27 This question has two parts. On January 1st, a person deposits 1000 dollars into an account. Each following year, the balance grows by a fixed factor, forming a geometric sequence, and the value at the start of year  $n$  is  $u_n = 1000(1.05)^{n-1}$ . (a) Find the balance at the start of year 6, to the nearest dollar. (b) Find the total of all deposits' grown values summed from year 1 through year 6 (i.e.  $S_6$ ), to the nearest dollar.